Do Now · retrieval

Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What property does binary search need that linear search does not?

Q2

What time complexity is typical of merge sort?

Q3

For n=5, how many times does a single for-loop from 0 to n-1 run?

Q4

A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Count what grows — Big O describes how work increases with input size, not the exact milliseconds on one computer.

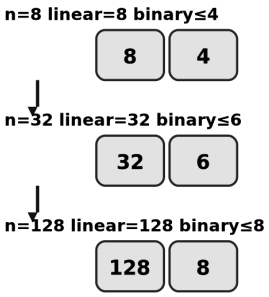
Core idea 2

Compare search — Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

Core idea 3

Compare sorting — A simple nested comparison loop can be $O(n^2)$, whereas merge sort is $O(n \log n)$.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

O(n^2) — Quadratic growth, often nested loops.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

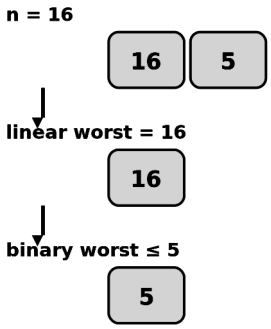
For 16 values, compare maximum comparisons of linear and binary search.

Follow the worked method

1. Predict first — Assume one successful or unsuccessful lookup in sorted data.

2. Model solution — Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
What happens when you double 32 to 64?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name the complexity of a single loop visiting n values.

Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Is binary search always better than linear search? Give one counterexample.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

O(log n) means execution time is literally log n seconds.

Use key vocabulary from memory

1. Explain time complexity in your own words.

2. Explain O(n) in your own words.

3. Explain O(log n) in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.